

force sensors can detect early equipment failures by monitoring vibrations. However, the deployment of printed sensors raises significant security and privacy issues, necessitating robust encryption and secure communication protocols to safeguard against cyber threats, especially in critical infrastructure.

Despite their advantages, printed sensors face considerable technical challenges in automotive systems, including integration with existing technologies and issues with reliability and durability. Rigorous optimisation of the printing process is necessary to achieve consistent performance. Continuous advancements in material science and manufacturing are crucial to addressing these challenges. Additionally, technical and security challenges must be overcome for the widespread adoption and success of these technologies in the IoT landscape. As industries innovate, the role of printed sensors is expected to expand, further driving advancements and opening new opportunities.

Sensitivity. Loss occurs because the printing process can reduce the effectiveness of the active sensing material.

Sensor drift. Deviation from baseline measurements is particularly problematic in applications where the sensor undergoes repeated deformations, such as in printed pressure sensors.

The flexibility of the sensors, while beneficial, can affect their reliability and the robustness of the sensor materials. Despite these advantages, printed sensors in IoT still face significant hurdles. The primary challenge is the technology's maturity and reliability compared to conventional sensors, often due to their proven performance and lower cost in standard applications. Additionally, while printed sensors offer multifunctionality, this feature is not always necessary for certain IoT applications, where simpler, dedicated sensors may suffice.

Encapsulation issues intensify

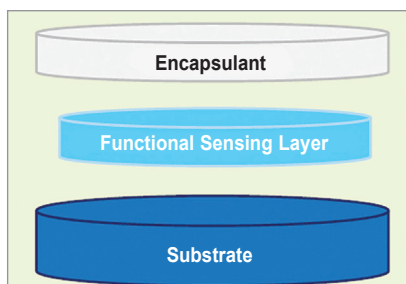


Fig. 3: Printed sensor architecture

printed sensors' challenges, where protective coatings may degrade, impacting performance. Innovations in sensor architecture and materials are underway to overcome these limitations, including designing more robust functional layers and improving ink formulations for better durability under stress. Enhancements in encapsulation technologies are also being made to protect sensors from environmental degradation.

To address these obstacles, companies focus on integrating printed sensors into broader applications, such as combining temperature and pressure sensors in a single module for electric vehicle battery monitoring. This integrated approach aims to enhance the appeal to OEMs and end-users, showcasing printed sensors' unique capabilities and improving market alignment.

Driving the future

Printed sensors are emerging as a critical innovation in the automotive sector, reshaping how vehicles are experienced and operated. Here are the key highlights:

Interactive interior surfaces.

Integrating conformal printed capacitive touch sensors transforms vehicle interiors into interactive hubs. These sensors allow drivers to control infotainment systems through simple touches or gestures, such as adjusting the volume or navigating menus.

Enhanced security with biometrics. Printed sensors enhance vehicle security by incorporating biomet-

ric fingerprint authentication. This technology, situated under display screens, authenticates drivers and provides personalised access, reducing the need for traditional keys.

Support for electric vehicles (EVs). Printed sensors are pivotal in developing advanced thermal management solutions essential for maintaining battery efficiency and vehicle performance. They contribute to reducing overall vehicle weight, further enhancing EVs' performance and range.

Advanced driver-assistance systems (ADAS). Integration with transparent conductive films in ADAS sensors boosts vehicle autonomy by improving passenger-vehicle interactions and enhancing safety features.

Comfort and safety innovations. Seat occupancy detection uses printed force sensors integrated with heaters to guarantee safety and comfort by accurately detecting if a seat is occupied. The sensors assess various factors such as strain, gas, pressure, and temperature to ensure the batteries operate efficiently and last longer.

Beyond individual vehicle enhancements, printed sensors offer a significant advantage in market differentiation. In an industry where many vehicles possess similar features, these sensors allow automakers to distinguish their products by offering advanced, personalised features that meet evolving consumer expectations. This innovation enhances vehicle functionality and plays a strategic role in positioning products in a competitive market. Printed sensors are setting the stage for the future of the automotive industry, emphasising safety, efficiency, and user experience. Their role is expected to expand as they continue to drive innovation in electric vehicles and beyond.

Regulating innovation: The impact of safety standards

Regulatory standards and safety certifications are critical for the Internet of